NARAYANA IIT ACADEMY INDIA

Sec: Sr. IIT **Race -11** **Date: 26-02-24**

PHYSICS

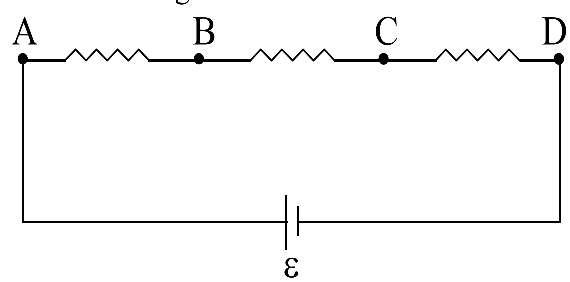
SECTION – I

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

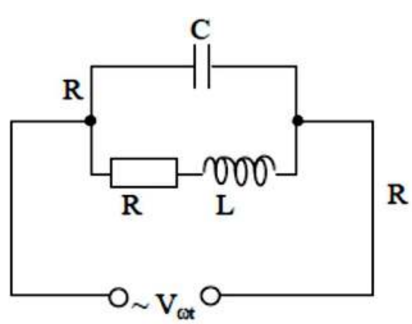
Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

1. Three identical resistances are connected in series with an ideal battery of emf ε=62V. A voltmeter having finite resistance is used to measure potential drops across different resistors. It is found that voltmeter reads $V_{AB} = V_{BC} = V_{CD} = 20$ volt. If same voltmeter is used to measure potential drop across different points then it's reading will be



A) $V_{AC} = 41\,\text{volt}$ B) $V_{AC} = 40\,\text{volt}$ C) $V_{AD} = 60\,\text{volt}$ D) $V_{AD} = 62\,\text{volt}$

2. If in the AC circuit shown, $R = 1\Omega, C = \dfrac{1}{80\pi}\text{F},\ L = \dfrac{1}{40\pi}\text{H}$ and, frequency of AC source=20 Hz, then select correct option(s).



A) Impedance of the AC circuit is 1Ω
B) Impedance of the AC circuit is 2Ω
C) Current through an AC source will lead its EMF
D) Current through an AC source will be in phase with its EMF

3. Mark the correct choices (x, y denotes x and y coordinates respectively and a is a constant of appropriate dimensions).

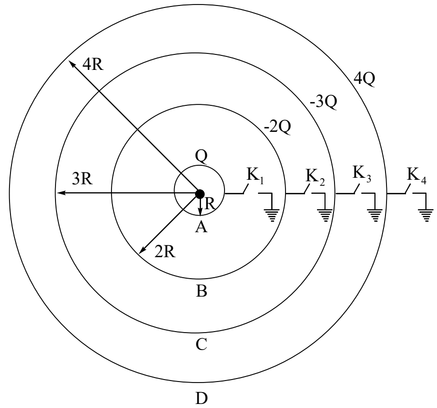
A) If electric field is given by $\vec{E} = ax\hat{i} + ay\hat{j}$, then shape of equipotential lines would be circular

B) If electric field is given by $\vec{E} = ay\hat{i} + ax\hat{j}$, then shape of equipotential lines would be circular.

C) If electric field is given by $\vec{E} = ay\hat{i} + ax\hat{j}$, then shape of equipotential lines would be rectangular hyperbola.

D) If $\vec{E} = ay\hat{i} - ax\hat{j}$, then we can't define electric potential for this field.

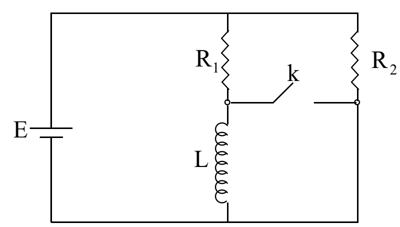
4. Four concentric conducting spherical shells of radii R, 2R, 3R and 4R carry charges Q, –2Q, –3Q and 4Q respectively. These shells are provided with switches K_1, K_2, K_3 and K_4 respectively. Which of the following operations will not change electric potential of spherical shell A?



A) switch K_1 alone is closed B) Both K_1 and K_2 are closed

C) Both K_2 and K_3 are closed D) K_4 alone is closed

5. In the electric circuit shown, the resistors R_1 and R_2 have unequal (non-zero) resistance and L is an ideal inductor. The cell is ideal and all conducting wires have zero resistance. Initially the key k was open and the circuit was in steady state. Then just after the key k is closed which of the following statement(s) is/are correct?



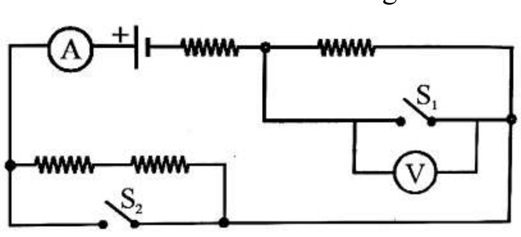
A) Current through R_1 will not change

B) Current through R_2 will not change

C) Current through L will not change

D) Current will pass through closed key

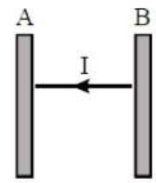
6. Shown is a DC circuit that contains two switches S_1 and S_2. Each switch has zero resistance when closed. All of the connecting wires should be considered to have zero resistance. All of the resistors shown are identical. The circuit contains an ideal ammeter and an ideal voltmeter. The diagram shows the switches open. Below the diagram are four different switch configurations for the circuit. Rank these configurations in terms of the ammeter reading & voltmeter reading.



Configuration	S_1	S_2
a	open	open
b	open	closed
c	closed	open
d	closed	closed

A) $i_a < i_b < i_c < i_d$ B) $i_a < i_c < i_b < i_d$ C) $v_c = v_d < v_a < v_b$ D) $v_c = v_d < v_b < v_a$

7. A parallel beam of light of intensity I is continuously incident normally on a plane surface A which absorbs 50% of the incident light and reflect remaining light in backward direction. The reflected light falls on B which is perfect reflector. The light reflected by B is again partly reflected and partly absorbed by A and this process continues. For all absorption by A, absorption coefficient remains equal to 0.5. The pressure experienced by A due to light is



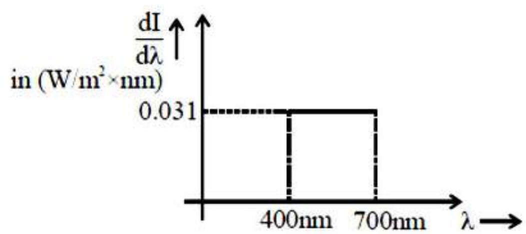
A) $\frac{1.5I}{c}$ B) $\frac{I}{c}$ C) $\frac{3I}{2c}$ D) $\frac{3I}{c}$

SECTION-II
(INTEGER ANSWER TYPE)

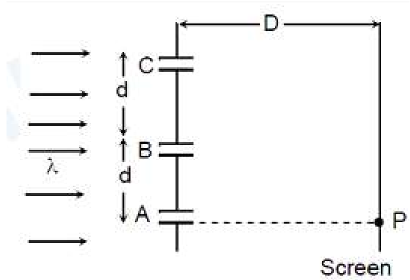
This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).
Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.

8. A charging device consists of eight identical, ideal cells in a proper series connection and a capacitor is charged using this device. Charging process is done by two ways : In first case, capacitor is connected across the whole device (eight cells) in a single step. In second case charging is done in steps which means, first capacitor is connected across a single cell, then across two cells, ..., then across eight cells. If loss of heat in first case is E_1 and in second case is E_2, then find $\frac{E_1}{E_2}$.

9. Scientists have made a light source whose spectral emissive power $\frac{dI}{d\lambda}$ is constant over visible range. Here I is intensity and λ is wavelength. In other words, $\frac{dI}{d\lambda}$ graph is shown below. This beam of area $1m^2$ is incident on emitter plate of photoelectric tube. The collector plate is sufficiently positive so that tube is in saturation mode. Assume that each capable photon liberates 1 electron. If work function is 2eV, what is the current (in A) in the tube? Round off to nearest integer.

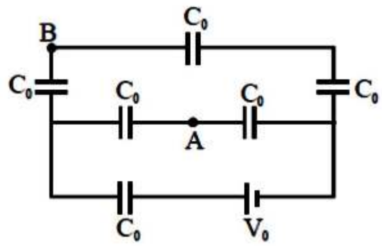


10. Three equidistant slits are illuminated by a monochromatic parallel beam of light falling normally onto them. The screen is parallel to the plane of the slits and the slit separation d << D (D is the slit-screen separation). The observation point P is directly opposite to A. It is given that BP – AP = $\frac{\lambda}{4}$, λ being the wavelength of light used. Find the ratio of the intensity at P when all slits are open to the intensity at P when only one slit is open.



11. Two equi-convex thin lenses 'A' and 'B' of focal lengths 20 cm and 30 cm respectively are placed at a separation 'd' with their principal axes coinciding. An object placed at a distance 'u' from the lens 'A' (away from B) forms an image at a distance 'v' from the lens 'B'(away from A). If it is observed that u and v have the same set of values for d = 30 cm, 60 cm, 120 cm and 150 cm and ratio $\frac{v}{u}=\frac{n}{4}$ then n=?

12. In the arrangement shown in figure, find the potential difference $V_B - V_A$ (in V). (Take $V_0 = 55$ V)



SECTION – III

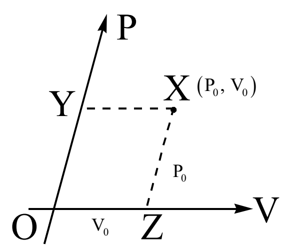
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

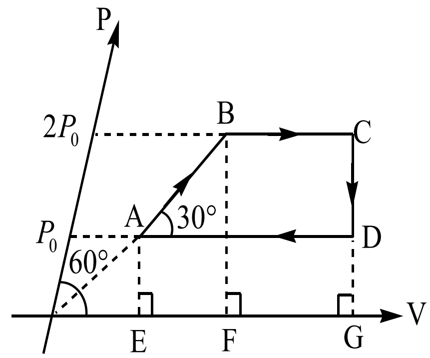
Answer the Q.No:13, 14 and 15 by appropriately matching the information given in the following table.

A thermodynamic reversible cyclic process ABCDA of one mole of an ideal mono-atomic gas $\left(C_V=\frac{3R}{2}\right)$ is drawn on a PV indicator diagram. The catch is that instead of usual orthogonal axes, the 'P' axis is inclined to 'V' axis at an angle $60°$. In an inclined axes system (say x-y), the co-ordinates of a point in first quadrant (x, y) should be understood as follows: From origin, x is the distance of the point along
x-axis, y is the distance of the point along y-axis. For example, the co-ordinate (P_0, V_0) is read on the diagram (graph) as shown



$XY = OZ = V_0$ and $OY = XZ = P_0$

Here scale on both P and V axes is 1 unit = 1 corresponding SI unit. $Q, \Delta V, \Delta T$ in this situation have usual meanings. β in a process is defined as ratio of work done by gas in a process to the modulus of the area of the PV graph of a process projected "normally" on to 'V' axis. For example in the cyclic processes ABCDA and for process AB, area referred to in the above statement is area of trapezium ABFE and for process BC, area is area of trapezium BCGF. Column I refers to a thermodynamic process, column II refers to a physical quantity and column III refers to their value. Process DA and BC are parallel to V-axis, process CD is perpendicular to V-axis and process AB if extended passes through origin and is inclined to V-axis by $30°$.

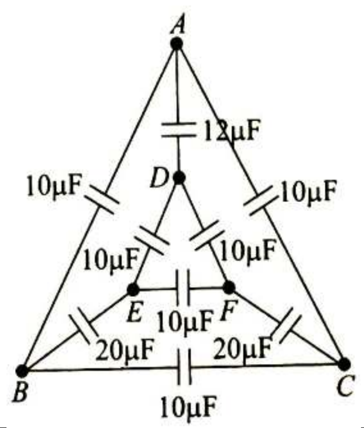


	Column I		Column II		Column III
(I)	AB	(i)	β	(P)	Positive
(II)	BC	(ii)	Q	(Q)	Negative
(III)	CD	(iii)	ΔV	(R)	Zero
(IV)	DA	(iv)	ΔT	(S)	1

13. Which one of the following options is the **CORRECT** combination?
A) (I) (i) (S) B) (II) (i) (Q) C) (IV) (i) (Q) D) (III) (iii) (R)
14. Which one of the following options is the **INCORRECT** combination?
A) (II) (ii) (P) B) (IV) (ii) (Q) C) (I) (i) (P) D) (III) (iv) (R)
15. Which one of the following options is the **INCORRECT** combination?
A) (I) (iv) (P) B) (II) (iv) (P) C) (IV) (i) (Q) D) (III) (i) (Q)

Answer the Q.No:16, 17 and 18 by appropriately matching the information given in the following table.

Answer the following three questions based upon the given diagram (take all capacitors to be uncharged initially)



Column I – Charge drawn from battery in μC	Column II – Charge in capacitor connected between A & D in μC	Column III – Charge in capacitor connected between B & C in μC
(I) 100	(i) 120	(P) 0
(II) 210	(ii) 0	(Q) 40
(III) 180	(iii) 40	(R) 50
(IV) 200	(iv) 50	(S) 100

16. If a 10 V battery is connected across the terminals A and D:
A) (I) (ii) (P) B) (IV) (i) (P) C) (II) (i) (Q) D) (IV) (iii) (S)
17. If a 10 V battery is connected across the terminals B and C:
A) (I) (ii) (P) B) (II) (iii) (R) C) (II) (ii) (S) D) (IV) (iii) (S)
18. If a 10 V battery is connected across the terminals E and F:
A) (I) (i) (P) B) (II) (ii) (Q) C) (III) (ii) (Q) D) (IV) (iii) (S)

CHEMISTRY
SECTION – I
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

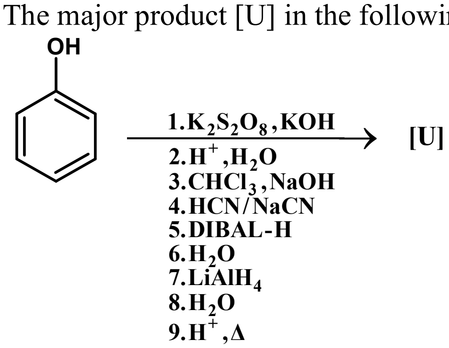
Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

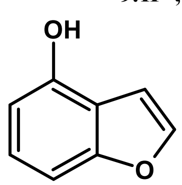
19. Find the correct statement(s)-
A) $CHCl_3$ undergoes alkaline hydrolysis much more rapidly than CCl_4 or CH_2Cl_2
B) Hydrolysis of chloroform is carried out in D_2O in presence of OD^-, when the reaction is interrupted and unconsumed chloroform is recovered, it is found to contain deuterium.
C) The presence of added Cl^- slows down alkaline hydrolysis of $CHCl_3$
D) When alkaline hydrolysis of $CHCl_3$ in presence of I^- is interrupted, there is recovered not only $CHCl_3$ but also $CHCl_2I$.

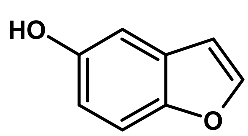
20. Micelle formation is accompanied by the
A) decrease in overall entropy due to ordering
B) increase in overall entropy due to increase in solvent entropy
C) increase in overall entropy due to increase in solute entropy
D) increase in overall entropy and decrease in enthalpy

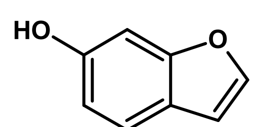
21. The initial rate of hydrolysis of methyl acetate (1M) by a weak acid (HA, 1M) is $\frac{1}{1000}$th of that of a strong acid (HX, 1M), at $25°C$. Then choose the **correct** statement(s):
A) K_a of HA is : 10^{-5}
B) p^{K_b} of A^- is : 8
C) pH of 0.1 M HA is : 3.5
D) pH of 0.1 M NaA is: 9.5

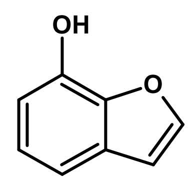
22. The major product [U] in the following reaction sequence is-



A) 

B) 

C) 

D) 

23. If the freezing point of 0.1 molal aqueous solution of cobalt (III) chloride - ammonia complex $(\alpha = 1)$ is $-0.558°C$. Then choose the **correct** statement(s):
$\left[\text{Given } K_f = 1.86\,K\,kg\,mol^{-1}, K_b = 0.52\,K\,kg\,mol^{-1}, R = 0.0821\,L\,atm\,mol^{-1}K^{-1}, m = M\right]$
A) The formula of the complex is $\left[Co(NH_3)_4Cl_2\right]Cl$
B) The bp of the aqueous solution is 100.156
C) The vapour pressure of the solution at $100°C$ is 721 mm of Hg
D) The osmotic pressure of the solution at $25°C$ is 7.34 atm

24. Which of the following option(s) represents the correct first - transition metal **M** that satisfies the requirements given?

A) $\left[M(H_2O)_6\right]^{3+}$ having one unpaired electron: Co

B) $\left[MBr_4\right]^{-}$ having the most unpaired electrons: Fe

C) diamagnetic $\left[M(CN)_4\right]^{3-}$: Cu

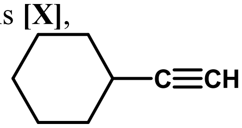
D) $\left[M(H_2O)_6\right]^{2+}$ with CFSE$=-\frac{3}{5}\Delta_0$: Cr

25. Which of the following is (are) dibasic acid(s) with bridging oxo group?
A) pyrosulphuric acid B) pyrophosphoric acid
C) pyrophosphorus acid D) pyrosulphurous acid

SECTION-II
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).
Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.

26. The total possible structure(s) (including stereomers) of R-X (mol. formula $C_8H_{11}Cl$) involved in the following reaction is **[X]**,

$\mathbf{R-X} \xrightarrow[\mathbf{2.H_2O}]{\mathbf{1.Mg,Et_2O}}$ 

Find [**X**-6].

27. Consider the electrochemical cell $M(s)\,|\,MI_2(sat)\,|\,M^{2+}(aq)\,|\,M(s)$ where 'M' is a metal. At 298K the standard reduction potentials are $E^o_{M^{2+}(aq)/M(s)} = -0.12V, E^o_{MI_2(sat)/M(s)} = -0.36V$ and the temperature coefficient is $1.1\times10^{-4}\ VK^{-1}$. At this temperature the standard enthalpy change for the overall reaction, ΔH^o_r, is -5**x** kJ/mole. Then the value of **x** is:
(1F = 96500C)

28. The $\Delta S^o_{universe}$ for the following reaction at 298K is approximately $\frac{y}{10}\, kJ\, K^{-1}\, mol^{-1}$. The value of **y** is:

$$H_2(g)+\frac{1}{2}O_2(g)\longrightarrow H_2O(\ell)$$

Given: $\Delta H^o_r = -285\, kJ\, mol^{-1}$, $S^o_{H_2O(\ell)} = 70\, JK^{-1}mol^{-1}$ $S^o_{O_2(g)} = 204\, JK^{-1}mol^{-1}$ $S^o_{H_2(g)} = 130\, JK^{-1}mol^{-1}$

29. How many of the following species have more bond length than C_2?
Li_2, B_2, N_2, O_2, F_2

30. How many of the following ores are subjected to froth floatation and then carbon reduction for extraction of corresponding metal? Chalcopyrite, Calamine, Sphalerite, Heamatite, Argentite, Cinnabar, Azurite, Galena, Cassiterite.

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer the Q 31, 32 and 33 by appropriately matching the information given in the following table.

Consider the cations given column-I, dry test(s) given in column-II and wet tests given in column-III.

Column-I	Column-II	Column-III
I) Co^{2+}	i) Bluish green colour in flame test.	P) White precipitate with KI.
II) Pb^{2+}	ii) Blue colour in flame test.	Q) Blue colour solution with NH_4SCN
III) Cu^{2+}	iii) Green coloured borax bead in reducing flame	R) Brown colour solution with

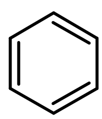
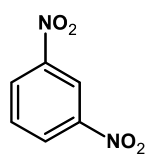
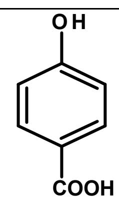
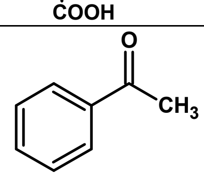
		$K_3[Fe(CN)_6]$
IV) Fe^{3+}	iv) Blue coloured borax bead in reducing flame	S) White precipitate with KCN which is insoluble in excess KCN.

31. Which of the following combination is correct for the ion which gives black precipitate with H_2S in presence of dil. HCl?
A) I, iv, Q B) II, ii, P C) III, i, P D) II, iv, P

32. Which of the following combination is correct for the ion which gives white precipitate with dil. HCl?
A) II, i, P B) II, ii, P C) II, ii, S D) II, i, S

33. Which of the following combination is correct for the ion which gives brown precipitatewith NH_4OH in presence of NH_4Cl?
A) IV, iv, R B) IV, iii, R C) III, ii, P D) I, iv, Q

Answer the Q34, 35 and 36 by appropriately matching the information given in the following table.

COLUMN - I		COLUMN II		COLUMN III	
A)		1)	$\xrightarrow[\text{ii) } Na_2S \text{ iii) } HNO_2 \text{ iv) } CuBr, \Delta]{\text{i) } HNO_3, H_2SO_4, (60^\circ)}$	P)	C_6H_6
B)		2)	i) NaOI ii) $H^{\oplus}$ (workup) iii) NaOH, CaO, Δ iv) CH_3Cl, $AlCl_3$ v) CrO_2Cl_2 / CS_2 vi) $H_3O^{\oplus}$	Q)	C_6H_6NF
C)		3)	$\xrightarrow[\text{iii) } HBF_4, \Delta \text{ iv) } Sn/HCl]{\text{i) } NH_3\text{-}H_2S \text{ ii) } NaNO_2, HCl}$	R)	C_7H_6O
D)		4)	$\xrightarrow[\text{iii) } Mg(THF) \text{ iv) } H_2O]{\text{i) } Br_2, H_2O \text{ ii) Zn dust}}$	S)	$C_6H_4NO_2Br$

34. Find the correct combination -
A) B-2-R B) A-3-Q C) D-2-P D) C-4-P

35. Identify the incorrect combination -
A) A-1-S B) B-3-Q C) D-2-R D) B-2-S

36. Column-I reactant [D]can not be distinguished from its correct matched product in column III by-
A) Tollen's Test B) Fehling test C) Haloform test D) Schiff test

MATHS
SECTION – I
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

37. If α is the root of the equation x – tan x = 3where $\alpha \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$; then which of the following is/are correct? (Where [.] denotes the greatest integer function and {.} fractional part function)

A) $\lim_{x\to\alpha^+}\left[\frac{\max\left(\tan x,\{x\}\right)}{x-3}\right]=1$ B) $\lim_{x\to\alpha^+}\left[\frac{\min\left(\tan x,\{x\}\right)}{x-3}\right]=1$

C) $\lim_{x\to\alpha^-}\left[\frac{\min\left(\tan x,\{x\}\right)}{x-3}\right]=0$ D) $\lim_{x\to\alpha^-}\left[\frac{\max\left(\tan x,\{x\}\right)}{\tan x}\right]=1$

38. Which of the following option(s) is/are incorrect?

A) $\int_0^{\pi/8}\left|\frac{\sin(8nx)+\cos(8nx)}{x}\right|dx < \frac{2\sqrt{2}}{\pi}\left(1+\frac{1}{2}+\frac{1}{3}+....+\frac{1}{n}\right)$

B) $\frac{2}{\ln 3} < \int_0^1 3^{x^{1/3}}dx < \int_0^1 4^{x^{1/3}}dx$

C) $\int_0^{\pi/6}\sqrt{\sin x}\left(8-3\sqrt{\sin x}\right)dx > \frac{8\pi}{9}$ D) $\frac{1}{4} < \int_{\pi/4}^{\pi/3}\frac{\tan x}{x}dx < \frac{1}{\sqrt{3}}$

39. Let a>0, b>0, c>0 and a + b + c = 6 then $\frac{(ab+1)^2}{b^2}+\frac{(bc+1)^2}{c^2}+\frac{(ca+1)^2}{a^2}$ may be equal to

A) $\frac{75}{4}$ B) 35 C) 15 D) 10

40. Let matrix $A=\begin{bmatrix}1&1&1\\1&1&1\\1&1&1\end{bmatrix}$ and $B=\begin{bmatrix}2&-1&-1\\-1&2&-1\\-1&-1&2\end{bmatrix}$, then –

A) $(A+B)^{2005}=A^{2005}+B^{2005}$ B) $(A-B)^{2005}=A^{2005}-B^{2005}$

C) $(3A+7B)^n=3^{n-1}\left(3^nA+7^nB\right)$ D) $(3A+7B)^n=3^{2n}A+3^n7^nB$

41. Which of the following is/are correct?

A) The point of intersection of $\overrightarrow{AB}$ and $\overrightarrow{CD}$ where A (4, 7, 8), B (–1, –2, 1), C (2, 3, 4) and D (1, 2, 5) is $\left(\frac{3}{2},\frac{5}{2},\frac{9}{2}\right)$

B) If (α,β,γ) be a point on the plane 2x + 6y + 15z = 7 then the least value of $\alpha^2+\beta^2+25\gamma^2$ is 7

C) The circumcenter of the triangle formed by the points (3, 2, –5), (–3, 8, –5) and (–3, 2, 1) is the point (–1, 4, –3)

D) The length of intercept on the line $\frac{x+1}{-1}=\frac{y-12}{5}=\frac{z-7}{2}$ by the surface $11x^2-5y^2+z^2=0$ is $\sqrt{30}$ units

42. A rectangle HOMF has sides HO=11 and OM=5. A triangle ABC has H as the intersection of the altitude, O the centre of the circumscribed circle, M the mid–point of BC, and F the foot of the altitude from A, then which of the following is/are true?

A) Perimeter of ΔABC is greater than 70 B) Area of ΔABC in integer

C) One side of ΔABC is rational D) All sides of ΔABC are less than 30

43. Let $S_n = \sum_{r=1}^{n}\left(\frac{r^4 + r^3 n + r^2 n^2 + 2n^4}{n^5}\right)$ and $T_n = \sum_{r=0}^{n-1}\left(\frac{r^4 + r^3 n + r^2 n^2 + 2n^4}{n^5}\right)$,

n = 1, 2, 3,…… then which of the following is/are true?

A) $T_n > \frac{167}{60}$ B) $T_n < \frac{167}{60}$ C) $S_n > \frac{167}{60}$ D) $S_n < \frac{167}{60}$

SECTION-II

(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.

44. If the number of ordered triplets (a, b, c) such that L.C.M (a, b) = 1000, L.C.M (b, c) = 2000 and L. C. M (c, a) = 2000 be N then sum of digits of N is/are ___________

45. The equation of a plane which passes through the line of intersection of the planes x + y + z – 1 = 0 and x – y + z – 3 = 0 and which is at maximum distance from the point (3, 4, 5) is ax + by + cz – d = 0, the value of $\left(\frac{a+b-c}{d}\right)$ is

46. If $f(x) = \begin{cases} e^x + e^{-x} & x < 0 \\ \log_{10}(100-x), x \in [0,100) \end{cases}$ then the number of solution of the equation $f(x) + \left(e - \frac{1}{e}\right)x = \frac{2}{e}$ is

47. Let $\frac{1}{a_1 - 2i}, \frac{1}{a_2 - 2i}, \frac{1}{a_3 - 2i}, \frac{1}{a_4 - 2i}, \frac{1}{a_5 - 2i}, \frac{1}{a_6 - 2i}, \frac{1}{a_7 - 2i}, \frac{1}{a_8 - 2i}$ are vertices of regular octagon. Then the area of octagon is (1/A)

(where $a_j \in R$ for j = 1,2,3,4,5,6,7,8 and $i = \sqrt{-1}$ where ($A^2/8$) is

48. The quadratic equation $2x^2 - 5\pi x + p = 0$ has roots equal to $2\sin^{-1}\alpha$ and $3\sin^{-1}\beta$, then $\left[\frac{p}{\pi}\right]$ equals, [.] denotes greatest integer function.

SECTION – III

(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.No:49, Q.No:50, Q.No:51 by appropriately matching the information given in the three columns of the following table.

Column 1, 2 & 3 contain parameters equation of conics, equation of chord joining two parametric point and equation of tangents at parametric point respectively

Column– 1	Column– 2	Column– 3
(I) $x=2\cos\theta$ $y=a\sin\theta$	(i) $\frac{x}{2}\cos\left(\frac{\theta_1-\theta_2}{2}\right)-\frac{y}{a}\sin\left(\frac{\theta_1+\theta_2}{2}\right)=\cos\left(\frac{\theta_1+\theta_2}{2}\right)$	(P) $\frac{x}{2}-\frac{y}{a}\sin\theta=\cos\theta$
(II) $x=2\sec\theta$ $y=a\tan\theta$	(ii) $\frac{x}{2}\sin\left(\frac{\theta_1+\theta_2}{2}\right)-\frac{y}{a}cos\left(\frac{\theta_1-\theta_2}{2}\right)+\cos\left(\frac{\theta_1+\theta_2}{2}\right)=0$	(Q) $y\sin 2\theta=2x\sin^2\theta+2a\cos^2\theta$
(III) $x=2\tan\theta$ $y=a\sec\theta$	(iii) $y\sin(\theta_1+\theta_2)=2x\sin\theta_1\sin\theta_2+2a\cos\theta_1\cos\theta_2$	(R) $\frac{x}{2}\cos\theta+\frac{y}{a}\sin\theta=1$
(IV) $x=a\cot^2\theta$ $y=2a\cot\theta$	(iv) $\frac{x}{2}\cos\left(\frac{\theta_1+\theta_2}{2}\right)+\frac{y}{a}\sin\left(\frac{\theta_1+\theta_2}{2}\right)=\cos\left(\frac{\theta_1-\theta_2}{2}\right)$	(S) $\frac{x}{2}\sin\theta-\frac{y}{a}+\cos\theta=0$

49. For a = 2 if a tangent is drawn to a suitable conic whose equation is x – 3y + 18 = 0, the correct combination with which the equation is obtained is:
A) (II) (i) (S) B) (III) (ii) (Q) C) (I) (ii) (R) D) (IV) (iii) (Q)

50. Which of the following is a correct combination?
A) (III) (i) (S) B) (II) (i) (P) C) (IV) (ii) (Q) D) (I) (ii) (R)

51. Which of the following is a correct combination?
A) (III) (ii) (S) B) (II) (iv) (P) C) (IV) (i) (Q) D) (I) (iii) (R)

Answer Q.No:52, Q.No:53, Q.No:54 by appropriately matching the information given in the three columns of the following table.

Column 1, 2 & 3 consists of different words, number of selection of 3 letters from the word and number of arrangements of 3 letters from the word respectively.

Column – 1	Column – 2	Column – 3
(I) DREAM	(i) 70	(P) 60
(II) DEDICATION	(ii) 10	(Q) 399
(III) POWERFUL	(iii) 77	(R) 336
(IV) COMBINATION	(iv) 56	(S) 378

52. Which of the following is a correct combination?
A) (I) (i) (P) B) (III) (iv) (R) C) (II) (ii) (S) D) (IV) (ii) (Q)

53. Which of the following is a correct combination?
A) (III) (iv) (S) B) (III) (i) (R) C) (IV) (iii) (Q) D) (I) (ii) (R)

54. Which of the following is correct combination?
A) (I) (ii) (S) B) (III) (i) (Q) C) (II) (i) (S) D) (IV) (iii) (R)

1	**BD**	2	**BD**	3	**ACD**	4	**ABD**	5	**ABC**
6	**BC**	7	**D**	8	**8**	9	**3**	10	**5**
11	**6**	12	**5**	13	**C**	14	**D**	15	**D**
16	**B**	17	**C**	18	**B**				
19	**ABCD**	20	**B**	21	**BCD**	22	**B**	23	**BD**
24	**BCD**	25	**AC**	26	**6**	27	**8**	28	**8**
29	**3**	30	**2**	31	**C**	32	**C**	33	**B**
34	**D**	35	**D**	36	**B**				
37	**ABCD**	38	**AC**	39	**AB**	40	**ABC**	41	**ACD**
42	**ABCD**	43	**BC**	44	**7**	45	**5**	46	**2**
47	**4**	48	**9**	49	**D**	50	**B**	51	**A**
52	**B**	53	**C**	54	**C**				